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Motor and motor controller thermal management features

Abstract

A system includes a gas turbine engine configured to provide propulsion to an aircraft and a starter system configured to start the gas turbine engine. The starter system comprises a motor controller and a closed-loop cooling system configured to cool the motor controller during an emergency in-flight restart operation of the gas turbine engine. The closed-loop cooling system includes a cooling fluid reservoir containing cooling fluid. The cooling fluid is configured to receive thermal energy from the motor controller during the emergency in-flight restart operation of the gas turbine engine.

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Background/Summary

TECHNICAL FIELD
(1) This disclosure relates to starting gas turbine engines.
BACKGROUND

(2) Turbine engines extract energy to perform work by compressing a working fluid, mixing a fuel into the compressed working fluid, igniting the fuel/fluid mixture, and expanding the combusted fuel/fluid mixture through a turbine. When a turbine is operating, a portion of the extracted energy is provided as the work input to the engine's compressor, thereby making the operation of the turbine self-sustaining. Prior to reaching this self-sustaining point, the work input to drive the compressor may be supplied by some system other than the turbine(s). These other systems often incorporate a starter that provides the motive force to turn the engine compressor, thereby providing an airflow to the turbine that can, eventually, extract and provide enough work output to drive the compressor.

SUMMARY

(3) Several different starter types may be used to start gas turbine engines. Some example gas turbine engine starter types include, air-turbine starters, electrical starters, hydraulic starters, and gas-cartridge starters. The gas turbine engine may be started by a controller executing an ignition sequence, where a spark is ignited in a combustor of the gas turbine engine when scheduled combustion conditions are reached (e.g., air flow, engine rotational speed, fuel flow, and the like.) Typically, a particular implementation may use only a single starter type (e.g., just one of an air-turbine starter or an electric starter, but not both). The starter type and ignition sequence may be chosen based on more or more factors including, for example, engine life, mechanical and/or thermal stresses, energy efficiency of the start, energy storage reduction (e.g., battery life reduction), noise restrictions, or the like. However, in some scenarios, an engine start may be desired as quickly and/or as reliably as possible, with less regard toward the potentially deleterious effects to these other considerations. Such scenarios may include emergency in-flight restarts, where at least one gas turbine engine of a plurality of gas turbine engines has been lost and needs to be restarted in the in-flight. Another such scenario, while unlikely, is an emergency in-flight restart where both engines are lost in a twin-engine aircraft, or each engine of a plurality of engines is lost.

(4) In such a situation, and in accordance with one or more aspects of this disclosure, an aircraft may be configured to perform an emergency in-flight restart. During an emergency in-flight restart, a controller may receive, either manually or automatically, an emergency restart command in response to loss of an engine while the aircraft is in-flight. Responsive to receiving the emergency restart command, the controller may determine whether at least a second gas turbine engine of a plurality of gas turbine engines is in operation. Responsive to receiving the emergency restart command and determining that at least the second gas turbine is in operation and that the first gas turbine is not in operation, the controller may cause execution of the emergency in-flight restart by at least simultaneously causing an air turbine starter and an electric machine to cause rotation of a spool of the gas turbine engine. In this way, the gas turbine engine may be restarted as quickly and reliably as possible by using energy from multiple sources, without regard or with only minimal regard to secondary considerations (e.g., energy efficiency, engine life, battery life, etc.). Performance of the emergency in-flight restart may help ensure that the aircraft may be safely landed, at which point the secondary considerations may be addressed.

(5) In some examples, the air turbine starter may receive fluid (e.g., air) from one or more sources external to the gas turbine during an emergency in-flight restart to cause additional rotation of the spool of the gas turbine engine. For example, bleed air from one or more additional gas turbine engines may be routed to the air turbine starter. In aircrafts which include an auxiliary power unit (APU), bleed air from the APU may be routed to the air-turbine starter, alone or in addition to bleed air from a second (operational) main gas turbine engine of a plurality of gas turbine engines. In some examples, a gas-cartridge or compressed air tank may supply further compressed air to the air-turbine starter. Supplying fluid from any one or more than one of these sources to an air turbine starter may cause rotation of the spool to restart the engine. Supplying fluid from a combination of sources may advantageously allow the spool to reach a threshold rotational speed for a start to be executed more quickly.

(6) In combination with multiple starting modes or all alone, other efforts to restart an engine in-flight may be employed. For example, an ignition sequence of the gas turbine engine may be modified for faster and/or easier restarting of the engine in-flight. For example, during a normal start, with an aircraft on the ground, fuel may be introduced to a combustor and combusted in the combustor at a first threshold rotational speed of the first spool, resulting in a first turbine temperature (e.g., turbine entry temperature (TET) or turbine inlet temperature (TIT)) of the gas turbine engine. During an emergency in-flight restart, the ignition sequence may be modified such that fuel is introduced and combusted at a second threshold rotational speed, resulting in a second turbine temperature. The second threshold rotational speed may be lower than the first threshold rotational speed, and the second turbine temperature may accordingly be higher. Although operation at the second turbine temperature may be deleterious to engine life over the long-term, modification of the ignition sequence in this way may result in execution of an emergency in-flight restart, which may help ensure that the aircraft may be safely landed.
(7) In examples which include using an electric machine controlled by a motor controller to cause rotation of a spool of the gas turbine engine, systems and techniques according to the present disclosure may manage thermal energy generated by the electric restart procedure in the starter system (e.g., the motor controller and electric machine (electric starter motor)). Absent such a cooling system, power switches in the motor controller may overheat and fail. The disclosed cooling systems and techniques may be closed-loop and/or lightweight, because the electric machine need only operate until the emergency in-flight restart is complete and operation of the gas turbine engine is self-sustaining. Since the emergency in-flight restart may only take seconds or minutes, a closed-loop cooling system may not include a heat exchanger and/or other equipment typically associated with thermal energy systems. Furthermore, the closed-loop cooling system may be sized to absorb a maximum amount of thermal energy that is substantially equal to the maximum amount of thermal energy that can be generated during the emergency in-flight restart. In this way, the electric machine may cause rotation of the spool during all or a portion of the emergency restart procedure without the motor controller overheating and while only adding minimal weight to the aircraft.
(8) In some examples, the disclosure is directed to a starting apparatus for a first gas turbine engine of a plurality of gas turbine engines of an aircraft. The apparatus includes an air turbine starter configured to cause rotation of a spool of the first gas turbine engine of the plurality of gas turbine engines using compressed air derived from a source external to the gas turbine engine. The apparatus further includes an electric machine configured to cause rotation of the spool of the first gas turbine engine. The apparatus includes a controller configured to: receive an emergency restart command for the first gas turbine engine while the aircraft is in-flight; determine whether the first gas turbine engine is in operation; determine whether at least a second gas turbine engine of the plurality of gas turbine engines is in operation; and responsive to receiving the emergency restart command and determining that at least the second gas turbine engine of the plurality of gas turbine engines is in operation and that the first gas turbine engine is not in operation, perform an emergency restart of the first gas turbine by at least simultaneously causing the air turbine starter and the electric machine to cause rotation of the spool of the first gas turbine engine.
(9) In some examples, the disclosure is directed to a method of starting a first gas turbine engine of a plurality of gas turbine engines in-flight. The method includes receiving, via a controller, an emergency restart command for the first gas turbine engine of the plurality of gas turbine engines while the aircraft is in-flight. The method includes determining, via the controller, whether the first gas turbine engine is in operation and, responsive to determining that the first gas turbine engine is not in operation, determining whether at least a second gas turbine engine of the plurality of gas turbine engines is in operation. The method includes, responsive to receiving the emergency restart command and determining that at least the second gas turbine engine of the plurality of gas turbine engines is in operation and that the first gas turbine engine is not in operation, performing an

emergency restart of the first gas turbine by at least simultaneously causing an air turbine starter and an electric machine to cause rotation of a spool of the first gas turbine engine.
(10) In some examples, the disclosure is directed to a starting apparatus for a gas turbine engine of a plurality of gas turbine engines of an aircraft. The apparatus includes a fuel supply system, a combustor, and a controller. The controller is configured to cause fuel to be introduced to the combustor of the gas turbine engine at a first threshold rotational speed of the gas-turbine engine during a normal starting operation in which the aircraft is on the ground. The controller is configured to cause fuel to be introduced to the combustor at a second threshold rotational speed of the gas-turbine engine during an emergency in-flight restarting operation in which the aircraft is in-flight. The second threshold rotational speed is lower than the first threshold rotational speed. Introducing fuel at the second threshold rotational speed results in a higher temperature in a turbine of the gas turbine engine than introducing fuel at the first threshold rotational speed.
(11) In some examples, the disclosure is directed to a starting method for a gas turbine engine of a plurality of gas turbine engines of an aircraft. The method includes causing, via control by a controller, fuel to be introduced from a fuel supply to a combustor of the gas turbine engine at a first threshold rotational speed during a normal starting operation in which the aircraft is on the ground. The method includes causing, via control by the controller, fuel to be introduced from the fuel supply to the combustor at a second threshold rotational speed during an emergency restarting operation in which the aircraft is in-flight. The second threshold rotational speed is lower than the first threshold rotational speed. Introducing fuel at the second threshold rotational speed results in a higher temperature in a turbine of the gas turbine engine.
(12) In some examples, the disclosure is directed to a system which includes a gas turbine engine configured to provide propulsion to an aircraft. The system further includes a starter system configured to start the gas turbine engine. The starter system includes a motor controller and a closed-loop cooling system configured to cool the motor controller during an emergency in-flight restart operation of the gas turbine engine. The closed-loop cooling system includes a cooling fluid reservoir containing cooling fluid, wherein the cooling fluid is configured to receive thermal energy from the motor controller during the emergency in-flight restart operation of the gas turbine engine.
(13) In some examples, the disclosure is directed to a method which includes starting a gas turbine engine configured to provide propulsion to an aircraft while the aircraft is in-flight by controlling a motor controller. The method further includes cooling the motor controller with a closed-loop cooling system during an emergency in-flight restart operation of the gas turbine engine. The closed-loop cooling system includes a cooling fluid reservoir containing cooling fluid. The cooling fluid is configured to receive thermal energy from the motor controller during the emergency in-flight restart operation of the gas turbine engine.
(14) The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a conceptual diagram illustrating an aircraft having multiple types of starters for gas turbine engines, in accordance with one or more aspects of this disclosure.
(2) FIG. **2** is a conceptual diagram illustrating further details of example gas turbine engines and associated systems, in accordance with one or more aspects of this disclosure.
(3) FIG. **3** is a conceptual diagram illustrating further details of other portions of the example aircraft of FIG. **2**, in accordance with one or more aspects of this disclosure.
(4) FIG. **4** is a flow diagram illustrating an example technique for restarting a gas turbine engine in-

flight, in accordance with one or more aspects of this disclosure.

(5) FIG. **5** is a flow diagram illustrating an example technique for restarting a gas turbine engine in-flight, in accordance with one or more aspects of this disclosure.

(6) FIG. **6** is a block diagram illustrating further details of example electric starters of gas turbine engines, in accordance with one or more aspects of this disclosure.

(7) FIG. **7** is a conceptual diagram illustrating the system of FIG. **2** with an example thermal energy management system.

(8) FIG. **8** is a conceptual diagram illustrating the system of FIG. **2** with another example thermal energy management system.

(9) FIG. **9** is a conceptual cross-section diagram of an example electric liquid-cooled electric machine housing according to the present disclosure.

(10) FIG. **10** is a conceptual cross-sectional diagram of an example air-cooled electric machine housing according to the present disclosure.

(11) FIG. **11** is a conceptual cross-sectional diagram of an example air-cooled electric machine housing according to the present disclosure that includes hollow fins.

(12) FIG. **12** is a flow diagram illustrating an example technique for managing thermal energy during an in-flight restart, in accordance with one or more examples of the present disclosure.

DETAILED DESCRIPTION

(13) FIG. **1** is a conceptual diagram illustrating an aircraft having multiple types of starters for gas turbine engines, in accordance with one or more aspects of this disclosure. As shown in FIG. **1**, system **100** may include aircraft **101** and one or more gas turbine engines **102**A and **102**B (collectively, “gas turbine engines **102**”) that drive one or more propulsors **104**A and **104**B (collectively, “propulsors **104**”). Aircraft **101** may be any type of aircraft including fixed wing, vertical takeoff and landing (VTOL), short takeoff and landing (STOL), rotorcraft, airplanes, and the like.

(14) Gas turbine engines **102** and propulsors **104** may collectively be configured to propel aircraft **101**. While illustrated in FIG. **1** in a turbo-prop configuration in which propulsors **104** are propellers, gas turbine engines **102** and propulsors **104** are not so limited. For instance, in other examples propulsors **104** may be fans such that gas turbine engines **102** and propulsors **104** may be in a turbo-fan configuration (e.g., low or high bypass). Furthermore, while illustrated in FIG. **1** as being wing-mounted, gas turbine engines **102** are not so limited. For instance, in other examples, gas turbine engines **102** and propulsors **104** may be fuselage mounted (e.g., mounted to a fuselage of aircraft **101** via pylons). In such examples, a twin-engine aircraft **101** may include first gas turbine engine **102**A disposed on a first side of fuselage **105** and second gas turbine engine **102**B mounted on a second opposite side.

(15) Aircraft **101** may include multiple starter types for each of gas turbine engines **102**. As one example, aircraft **101** may include both an air-turbine starter and an electric starter for each of gas turbine engines **102**. For instance, aircraft **101** may include a first air-turbine starter and a first electric starter for gas turbine engine **102**A and a second air-turbine starter and a second electric starter for gas turbine engine **102**B. Gas turbine engines **102**A and **102**B may be functionally substantially equivalent and/or interchangeable, such that description of first gas turbine engine **102**A may also apply to gas turbine engine **102**B. Aircraft **101** may include additional or different starter types. For instance, aircraft **101** may include two or more of air-turbine starters, electrical starters, hydraulic starters, and gas-cartridge starters. Each starter type may be configured to cause rotation of a spool of the gas turbine engine, such that gas turbine engines **102** may be rotated by one or more starters until enough work can be extracted such that the engine becomes self-sustaining.

(16) Including multiple starter types may present various advantages. As one example, including starters that utilize different energy sources may provide improved reliability in the event an energy source is not available. For instance, an aircraft **101** that includes air-turbine starters that use

compressed air and electric starters that use electrical energy may enable starting when either electrical energy or compressed air is available, but may not require both electrical energy and compressed air to be available. In some examples, when both compressed air and electrical energy are available, both types of starters may be employed to execute an emergency in-flight restart, which may be performed when one or both of gas turbine engines **102** are lost (e.g., become non-operational) in the air. Using both types of starters to execute an emergency in-flight restart may increase the speed and/or reliability of the in-flight restart.

(17) An air-turbine starter may need a source of compressed air to operate. In example aircrafts which include a plurality of gas turbine engines, bleed air from second gas turbine engine **102**B may supply compressed air to first gas turbine engine **102**A. In example aircrafts which include an auxiliary power unit (APU) (not illustrated in FIG. **1**) (e.g., a non-propulsion gas turbine engine), bleed air from the APU may provide compressed air to an air turbine starter. In the unlikely event of engine loss of gas turbine engine **102**A, gas turbine engine **102**B, or both (e.g., complete engine loss) while aircraft **101** is in flight, it may be desirable to start the lost gas turbine engine or gas turbine engines as reliably, quickly, and or safely as possible so that the engine can be restarted in-flight. For purposes of this disclosure, any in-air start of gas turbine engine **102**A and/or gas turbine **102**B may be called an emergency in-flight restart.

(18) In accordance with one or more aspects of this disclosure, aircraft **101** may be configured to simultaneously cause rotation of a spool of gas turbine engine **102**A during an emergency in-flight restart with energy from multiple starter types. In this way, the probability that a successful start is achieved may be increased, and the speed with which a successful start is achieved may also be increased. The multiple types of starters may include and electric starter and an air-turbine starter. In some examples, compressed air may be supplied to the air-turbine starter from a source external to gas turbine engine **102**A, such as from second gas turbine **102**B. Additionally, or in situations where both engines **102**A and **102**B must be restarted in-flight, compressed air may be supplied to the air-turbine starter from an APU, or from a compressed air tank. In some examples, compressed air may be supplied from more than one or the above sources, such as a first source and a second source, such that additional rotation (e.g., torque) may be generated by the air-turbine starter, which may start gas turbine engine **102**A more reliably and/or more quickly. In some examples, both the electric starter and air turbine starter may be configured to cause rotation of HP spool **220**A throughout the start (i.e., until gas turbine engine **102**A is self-sustaining). Causing rotation of the spool may, in some examples, help avoid a hung start, which is a condition where gas turbine engine **102**A lights properly but does not accelerate toward idling speed, but rather the rotational speed of the spool plateaus or drops off after fuel is injected and combustion begins.

(19) In accordance with one or more aspects of the disclosure, instead of or in addition to simultaneously causing rotation of a spool of gas turbine engine **102**A with both an electric starter and an air-turbine starter, the ignition sequence of a start procedure may be modified to more reliably and/or more quickly start gas turbine engine **102**A. Engine **102**A may include a fuel supply configured to feed a combustor, where the fuel is combusted with compressed air. Gas turbine engine **102**A may include a controller that is configured to cause fuel to be introduced and combusted when the spool reaches a first threshold rotational speed when aircraft **101** is on the ground and is performing a normal starting procedure. During and emergency in-flight restart, the controller may be configured to introduce fuel to the combustor at a second threshold rotational speed. The second threshold rotational speed may be lower than the first threshold rotational speed, such that fuel is introduced earlier during an emergency in-flight restart than it is during a normal ground start. Introduction of fuel at the second threshold rotational speed may result in a higher temperature in the combustor (which may be measured at the turbine and called the turbine entry temperature (TET) or turbine inlet temperature (TIT)) than introduction of fuel at the first threshold rotational speed. Put differently, introducing fuel earlier in the spool-up process may cause gas turbine engine **102**A to run hotter. Ordinarily, running the engine hotter may not be desirable

because the turbine temperature may approach or exceed a critical temperature, where component life is adversely impacted and/or maintenance procedures may be required up landing to inspect components for signs of overheat damage. However, the interest of reliably and/or quickly restarting engine **102**A during the emergency in-flight restart may outweigh these long-term costs, and the fuel schedule may be modified to make the restart faster and/or easier, so that aircraft **101** may be safely landed with operating gas turbine engines **102**.

(20) As mentioned above, and in accordance with one or more aspects of this disclosure, the electric starter may impart torque throughout the emergency in-flight restart operation, until available power is expended or gas turbine engine **102**A becomes self-sustaining. The emergency in-flight restart may take several tries and/or extend over a period of time from seconds to minutes. For example, the emergency in-flight restart may take from about one minute to about 15 minutes, or from about two minutes to about 10 minutes, or from about three minutes to about seven minutes. Accordingly, a starter system including an electric starter and associated motor controller(s) may generate thermal energy throughout the duration of the emergency in-flight restart. In some examples, the motor controller may have power switches which generate a power signal to control an electric starter motor which causes rotation of a spool of the gas turbine engine. These power switches may generate thermal energy and be prone to overheat during the emergency in-flight restart, where the electric motor is running continuously for the duration of the process. Systems and techniques according to the present disclosure may include a closed-loop cooling system configured to cool the motor controller during the emergency in-flight restart. In some examples, the closed-loop cooling system may include a cooling fluid reservoir and cooling fluid, but not a heat exchanger. Furthermore, the closed-loop cooling system may be sized to only cool for the duration of the emergency in-flight restart. In other words, in some examples, the closed loop cooling system may include enough cooling fluid to absorb a maximum amount of thermal energy, and the maximum amount of thermal energy may be substantially equal to the maximum amount of thermal energy that can be generated by the motor controller and/or electric starters during the emergency in-flight restart. In this way, the closed-loop cooling system may receive thermal energy from the motor controller during the emergency in-flight restart to keep the power switches from overheating, but not unnecessarily weigh down aircraft **101** with over-sized and/or unnecessary equipment.

(21) FIG. **2** is a conceptual diagram illustrating further details of example gas turbine engines and associated starters, in accordance with one or more aspects of this disclosure. System **200** of FIG. **2** may be an example of system **100** of FIG. **1**. As shown in FIG. **2**, system **200** may include gas turbine engines **202**A and **202**B (collectively, “gas turbine engines **202**”) and propulsors **204**A and **204**B (collectively, “propulsors **204**”), pneumatic distribution **205**, engine controllers **203**A and **203**B (collectively, “engine controllers **203**”), valves **223**A and **223**B (collectively, “valves **223**”), controller **206**, and electrical distribution **207**. Gas turbine engines **202** and propulsors **204** may be examples of gas turbine engines **102** and propulsors **104**. In some examples, propulsors **204** may be propellers, as illustrated in FIG. **2**.

(22) Gas turbine engines **202** may each include a compressors of compressors **210**A/**210**B (collectively, “compressors **210**”), a combustor of combustors **212**A/**212**B (collectively, “combustors **212**”), a turbine of turbines **214**A/**214**B (collectively, “turbines **214**”), a power turbine of power turbines **216**A/**216**B (collectively, “power turbines **216**”), a low-pressure (LP) spool of LP spools **218**A/**218**B (collectively, “LP spools **218**”), a high-pressure spool of HP spools **220**A/**220**B (collectively, “HP spools **220**”), a spool speed sensor of spool speed sensors **221**A/**221**B (collectively, “spool speed sensors **221**”), an air-turbine starter of air-turbine starters **222**A/**222**B (collectively, “air-turbine starters **222**”), an electric starter of electric starters **224**A/**224**B (collectively, “electric starters **224**”), and starter connection components of starter connection components **230**A/**230**B (collectively, “starter connection components **230**”).

(23) In operation, compressors **210** may compress air, combustors **212** may add fuel to the

compressed air and combust the resulting mixture, exhaust from the combustion may rotate turbines **214** and power turbines **216**. Turbines **214** and compressors **210** may be coupled to HP spools **220** (e.g., HP shafts). As such, rotation of turbines **214** result in rotation of compressors **210**. Power turbines **216** and propulsors **204** may be connected to LP spools **218** (e.g., LP shafts). As such, rotation of power turbines **216** may result in rotation of propulsors **204**. While described and illustrated as a free-power turbine (i.e., where propulsors **104** are propellers driven by power turbines **216**, and no compressor is attached to LP spools **218**, or a boost compressor attached to LP spools **218**), power turbines **216** are not so limited. For instance, in some examples, gas turbine engines **202** may include LP compressors coupled to LP spools **218**. Similarly, in some examples, gas turbine engines **202** may be three-spool engines.

(24) Spool speed sensors **221** may each be configured to measure a rotational speed of one or more spools (e.g., HP spools, LP spools, intermediate spools, etc.). For instance, spool speed sensor **221**A may be configured to measure a rotational speed of HP spool **220**A and spool speed sensor **221**B may be configured to measure a rotational speed of HP spool **220**B. Spool speed sensors **221** may be configured to output a representation of the measured rotational speeds to one or more other components of system **200**, such as EECs **203**. For instance, spool speed sensor **221**A may output a representation (e.g., digital or analog) of the rotational speed of HP spool **220**A to EEC **203**A and spool speed sensor **221**B may output a representation of the rotational speed of HP spool **220**B to EEC **203**B. In some examples, spool speed sensors **221** may each contain a temperature probe configured to sense a turbine temperature (TET and/or TIT), and likewise output a representation of the measured temperature to EEC **203**A. In such examples, spool speed sensors **221** may be combination speed/temperature sensors. In other examples, temperature may be measured by a discrete temperature sensor located elsewhere in gas turbine engine **202**A.

(25) Air-turbine starters (ATSs) **222** may be configured to rotate spools of gas turbine engines **202** to start gas turbine engines **202**. ATSs **222** may each include a turbine which may generate rotational force when compressed air is expanded across the turbine. Starter connection components **230** carry the resulting rotational force to the appropriate starting spool. For instance, starter connection components **230**A may include gears/shafts/etc. configured to carry rotational force from ATS **222**A to HP spool **220**A such that rotation of the turbine of ATS **222**A results in rotation of HP spool **220**A.

(26) Electric starters **224** may be configured to rotate spools of gas turbine engines **202** to start gas turbine engines **202**. Electric starters **224** may each include an electric machine which may generate rotational force using electrical energy. Examples of electric machines that may be included in electric starters **224** include, but are not limited to, alternators, dynamos, permanent magnet machines, field wound machines, synchronous, asynchronous, brushed, brushless, etc. Starter connection components **230** carry the resulting rotational force to the appropriate starting spool. For instance, starter connection components **230**A may include gears/shafts/etc. configured to carry rotational force from electric starter **224**A to HP spool **220**A such that rotation of the electric machine of electric starter **224**A results in rotation of HP spool **220**A.

(27) As noted above, starter connection components **230** may include components configured to convey rotational force generated by starters (e.g., ATSs **222** and/or electric starters **224**) to starting spools (e.g., HP spools **220**) of gas turbine engines **202**. As illustrated in the example of FIG. **2**, starter connection components **230** may include a tower shaft connected to a gearbox that joins the starters. However, it is understood that other arrangements of starter connection components **230** are contemplated. In some examples, both ATSs **222** and electric starters **224** may be mechanically coupled to the same start gear, which is not specifically illustrated within starter connection components **230**. The start gear may be a dual-ended gear comprising a first end and a second opposite end, and both ATSs **222** and electrical starters **224** may be connected to the first end. In this way, mechanical complexity may be reduced while allowing for multiple starting modes to cause rotation of HP spool **220**A.

(28) In some examples, starter connection components **230** may include one more clutches configured to rotationally de-couple the starters from the starting spools. For instance, starter connection components **230**A may include a clutch configured to rotationally decouple electric starter **224**A from HP spool **220**A. Inclusion of such a clutch may be desirable in that electric starter **224**A may be rotationally de-coupled from gas turbine engine **202**A. For instance, in the event of a fault in the electric machine of electric starter **224**A, the clutch may be actuated to rotationally de-couple electric starter **224**A from HP spool **220**A such that a rotor of the electric machine ceases to be driven, thereby mitigating the fault. However, inclusion of such a clutch may present one or more disadvantages. For instance, inclusion of the clutch may add additional weight (e.g., 10, 15, 20 lbs.) per engine of gas turbine engines **202**. As noted above, additional weight in the aerospace context is undesirable.
(29) Pneumatic distribution **205** may include various sources of compressed air and components configured to transport the compressed air about the aircraft. Example sources of compressed air include, but are not limited to, bleed air from an operating engine of gas turbine engine **202**, bleed air from the APU, compressed air storage tanks, and ground sources (e.g., huffer-cart). Pneumatic distribution **205** may provide the compressed air to various components of system **200**. For instance, pneumatic distribution **205** may provide compressed air to ATSs **222** (e.g., as controlled by valves **223**).
(30) Valves **223** may control the flow of compressed air to ATSs **222**. For instance, valve **223**A may selectively allow compressed air to flow from pneumatic distribution **205** to ATS **222**A and valve **223**B may selectively allow compressed air to flow from pneumatic distribution **205** to ATS **222**B. Valves **223** may control the flow of compressed air based on control signals received from other components of system **200**, such as EECs **203**. For instance, responsive to a signal received from EEC **203**A, valve **223**A may allow compressed air to flow from pneumatic distribution **205** to ATS **222**A.
(31) Electrical distribution **207** may include various sources of electrical power and components to transport the electrical power about the aircraft. Example sources of electrical power include, but are not limited to, an APU (e.g., a generator of the APU), generators of gas turbine engines **202**, ground sources (e.g., ground grid connection or ground power cart), batteries, or other on-board electrical systems. Electrical distribution **207** may provide the electrical power to various components of system **200**. For instance, electrical distribution **207** may provide electrical energy to controller **206**.
(32) Controller **206** may be configured to control operation of various components of system **200**, such as electric starters **224**. In some examples, controller **206** may be referred to as a common controller or a common electric motor controller in that it may control electric starters of a plurality of gas turbine engines **202**. For instance, controller **206** may control both electric starter **224**A and electric starter **224**B. As discussed in further detail below, controller **206** may include one or more power switches configured to generate power signals to drive electric starters **224** and/or one or more multiplexors.
(33) Engine controllers **203** may be configured to control various functions of gas turbine engines **202**. Engine controllers **203** of any of the above aspects may be implemented as a single controller or multiple separate (e.g., distributed) controllers. Engine controllers **203** may include one or more discrete controllers for each of gas turbine engines **202**. For instance, as shown in FIG. **2**, EEC **203**A may control functions of gas turbine engine **202**A and EEC **203**B may control functions of gas turbine engine **202**B. Thus, engine controllers **203** may be or may form part of a control system. Engine controllers **203** may be implemented in software, hardware or a combination of the two. Engine controllers **203** may be or may be a functional module of an Engine Electronic Controller (EEC) or a Full Authority Digital Engine Controller (FADEC). Although various functions are attributed herein to controller **206** and other functions are attributed herein to EECs **203**, it is understood that this is merely for ease of description, and controller **206** and EECs **203**

may have different functionality, or may be integrated into a single unit which performs all of the described functions.

(34) In a normal ground starting operation, engine controllers **203** may perform one or more operations to start gas turbine engines **202**. For instance, EEC **203**A may cause valve **223**A to open, thereby allowing compressed air to flow from pneumatic distribution **205** to ATS **222**A. The compressed air may expand across ATS **222**A, causing rotation of the turbine of ATS **222**A and thereby causing rotation of HP spool **220**A. EEC **203**A may monitor a rotational speed of HP spool **220**A (e.g., via spool speed sensor **221**A). Once the rotational speed of HP spool **220**A reaches a threshold rotational speed for combustion, EEC **203**A may cause fuel to be introduced into combustor **212**A and cause firing of ignitors within combustor **212**A. Once the rotational speed of HP spool **220**A reaches a self-sustaining threshold rotational speed, gas turbine engine **202**A may be self-sustaining and EEC **203**A may cause valve **223**A to close. EEC **203**B may perform similar techniques to start gas turbine engine **202**B.

(35) During an emergency in-flight restart (e.g., a mid-air restart), controller **206** may receive an emergency restart command for gas turbine engine **202**A from EEC **203**A. For example, the emergency restart command may be automatically generated when the rotational speed of HP spool **220**A or another spool of engine **202**A falls below a threshold level (e.g., 10% of idle speed, 20% of idle speed, 50% of idle speed, or the like), as sensed by spool speed sensor **221**A. Alternatively, controller **206** may receive the emergency restart command from the pilot manually. Responsive to receiving the emergency restart command, controller **206** may be configured to determine whether gas turbine **202**A is operational, based on the turbine temperature and/or rotational speed at speed sensor **221**A, or by another method such as the power level generated by power turbine **216**A. Responsive to determining that gas turbine engine **202**A is not in operation, controller **206** may determine whether at least second gas turbine engine **202**B or APU (**202**C, FIG. **3**) is in operation. Responsive to determining that gas turbine engine **202**B, APU (**202**C, FIG. **3**), or both is in operation, controller **206** may be configured to perform the emergency in-flight restart by at least simultaneously causing ATS **222**A and electric starter **224**A to cause rotation of HP spool **220**A of gas turbine engine **202**A.

(36) Both ATS **222**A and electric starter **224** may be utilized to restart gas turbine engines **202**A when gas turbine engine **202**A is not in operation. For instance, EEC **203**A may output a signal to controller **206** that causes controller **206** to utilize electrical energy from electrical distribution **207** to generate and output power signals to electric starter **224**A. The power signals may cause a magnetic field in electric starter **224**A that causes a rotor of electric starter **224**A to rotate. As noted above, rotational energy of the rotor may be carried to HP spool **220**A via starter connection components **230**. Compressed air may be supplied to ATS **222**A to assist the restart. For example, EEC **203**A may cause valve **223**A to open, thereby allowing compressed air to flow from pneumatic distribution **205** to ATS **222**A. The resulting rotation of HP spool **220**A, due to electric starter **224**A alone or in combination with rotational force imparted by ATS **222**A, may cause a rotational speed of HP spool **220**A to reach a level at which fuel can be re-introduced into combustor **212**A, thereby enabling a restart (e.g., re-light) of gas turbine engine **202**A.

(37) In examples where gas turbine engine **202**B has been lost, EEC **203**B may perform similar techniques to start gas turbine engine **202**B. In examples where both gas turbine engine **202**A and gas turbine engine **202**B have been lost, controller **206** may be configured to perform an emergency in-flight restart of engines **202** in series, or perform an emergency in-flight restart of engines **202** in parallel. In some examples, engine controllers **203** and/or controller **206** may automatically determine whether a series or parallel restart should occur based on the compressed air available from pneumatic distribution **205**. For example, if the compressed air available is below a threshold, such as in some versions of aircraft **101** when compressed air from only APU (**202**C, FIG. **3**) is available, controller **206** may determine that a series restart should occur. In a series restart, all of the available compressed air from pneumatic distribution **205** may be directed to ATS **222**A by

EEC **203**A causing valve **223**A to open and EEC **203**B causing valve **223**B to close or remain closed. In some examples, should the re-light of gas turbine engine **202**A be successful, EEC **203**B may then utilize ATS **222**B and controller **206** may cause electric starter **224**B to restart gas turbine engine **202**B. In some examples, gas turbines engines **202**A and **202**B may be started simultaneously (i.e., in parallel) by motor controllers **203** opening valves **223** and controller **206** causing electric starters **224** to cause rotation of HP spools **220**, such that compressed air from pneumatic distribution **205** flows to all ATSs **222** simultaneously (e.g., at the same time or substantially the same time), and/or electrical energy from electrical distribution **207** flows to all electric starters **224** simultaneously.

(38) With respect to examples where both gas turbine engines **202** are lost and controller **206** determines that gas turbine engines **202** should be restarted in series, while gas turbine engine **202**A is described as being restarted first, it should be understood that, in some examples, gas turbine engine **202**B may be restarted first. In other examples, only one of gas turbine engines **202** may be restarted (e.g., in the event that damage to one of gas turbine engines **202** is known).

(39) In some examples, EECs **203** may automatically determine which starters to use to start or restart gas turbine engines **202**. As one example, responsive to determining that at least a second gas turbine (e.g., propulsive, or non-propulsive) is in operation one compressed air source of pneumatic distribution **205** is available, engine controllers **203** may automatically determine to utilize at least ATS **222** to start gas turbine engines **202**. In some examples, engine controllers **203** may receive external signals (e.g., from pilots of aircraft **101**) indicating which starts to use to start or restart gas turbine engines **202**.

(40) As noted above, gas turbine engines **202** may be configured such that shafts that drive propulsors **204** (i.e., LP spools **218**) may not be connected to any compressors. As such, a windmilling emergency in-flight restart may not be possible as rotation of propulsors **204** may not result in much additional airflow through cores of gas turbine engines **202** (e.g., through combustors **212** and turbines **214**). As such, aspects of this disclosure related to emergency in-flight restarts using electric starters **224** and ATSs **222** may be particular useful in such arrangements. However, in some examples windmilling (e.g., allowing propulsors **204** to rotate freely) during an emergency in-flight restart may be helpful as another source of compressed air, however small, routed through the core. In some examples, especially where LP spools **218** are configured to drive propulsors **204**, system **200** may be configured such that a windmilling operation of propulsor **204**A drives a generator (not illustrated in FIG. **2**) that generates electrical power from the windmilling operation. The generated electrical power may be supplied to electric motors **224** for performing the emergency in-flight restart of gas turbine engines **202**. Using a windmilling operation to generate electrical power rather than airflow through the core may be more efficient, in some examples, and thus more helpful during the emergency in-flight restart. Furthermore, generating electrical energy from a windmilling operation may allow for reduced battery size (in examples where battery power is used by electrical distribution **207** to start electric motors **224**). In this way, gas turbine engine **202**A may generate at least a portion of the electrical energy used by electric starter **224**A to perform the emergency in-flight restart of gas turbine engine **202**A by EEC **203**A allowing propulsor **204**A to windmill as aircraft (**101**, FIG. **1**) moves through the air. As such, techniques of this disclosure may enable windmill re-start of a turbo-prop engine with a free power turbine. In some examples, such as in the case of a dual-engine flameout, mechanical and/or electrical energy generated by windmilling operation of both first gas turbine engine **202**A and second gas turbine engine **202**B may be simultaneously directed to first gas turbine engine **202**A either mechanically or electrically, as described above. In this way, another source of energy may contribute to reliably and/or quickly restarting first gas turbine **202**A.

(41) In some examples, where propulsors **204** are variable pitch propulsors, EECs **203** may be configured to manipulate the pitch of blades of propulsors **204** to increase the reliability and/or speed of the emergency in-flight restart. For example, EECs **203** may command feathering (i.e.,

rotating blades until they are edgewise to the flight direction, reducing drag) and/or unfeathering (rotating blades to be more perpendicular to the flight direction, increasing drag) of propulsors **204**. In some examples, during an emergency in-flight restart where propulsors **204** are windmilling, EECs **203** may command an unfeathering of the propulsor. In some examples, such an action may increase the amount of electrical energy generated by the windmilling operation, which may be supplied to electric starter **224** by electrical distribution **207**.

(42) As mentioned above, and in accordance with one or more aspects of this disclosure, ATS **222**A and/or electric starter **224**A may be configured to cause rotation of HP spool **220**A throughout the emergency in-flight restart. For example, electric starter **224**A may impart torque throughout the emergency in-flight restart process, until available power is expended or gas turbine engine **102**A becomes self-sustaining. The emergency in-flight restart may take several tries and/or extend over a period of time from seconds to minutes. For example, the emergency in-flight restart occur during a time period with duration of from about one minute to about 15 minutes, or from about two minutes to about 10 minutes, or from about three minutes to about seven minutes.

(43) FIG. **3** is a conceptual diagram illustrating further details of other portions of system **200** of FIG. **2**, in accordance with one or more aspects of this disclosure. FIG. **3** omits second gas turbine engine **202**B for clarity, and illustrates APU **202**C and compressed air tank **242** instead, although all of first gas turbine engine **202**A, second gas turbine engine **202**B, and APU **202**C are components of system **200**. APU **202**C may be a non-propulsive gas turbine engine, and generally may be smaller than main gas turbine engines **202**A and **202**B. APU **202**C may include electric starter **224**C, compressor **210**C, combustor **212**C, and turbine **214**C, which may be all disposed on shaft **218**C and configured to operate similarly to corresponding portions of main gas turbine engine **202** through control by EEC **203**C. However, rather than generating propulsion, in operation APU **202**C may be configured to generate electrical power by generator **242**C. The generated electrical power may be supplied to electrical distribution **207** for distribution. In some examples, APU **202**C may be located in a tail-section of a fuselage of aircraft **101** (FIG. **1**). In some examples, compressor **210**C of APU **202**C may be configured to compress air, and a bleed air line which includes valve **223**C. Valve **223**C may be opened to route bleed air from APU **202**C to pneumatic distribution **205**.

(44) System **200** may also include one or more compressed air tanks **242**, which may be tanks, gas cartridges, or other units configured to store compressed fluid (e.g., air) for routing to ATSs **222** during an emergency in-flight restart. For example, controller **206** may be configured to cause valve **223**D to open to route compressed air from compressed air tank **242** to pneumatic distribution **205**, where the compressed air may be routed to ATS **222**A to cause additional rotation of HP shaft **220**A during an emergency in-flight restart.

(45) In examples where EEC **203**A determines that gas turbine engine **202**A is not in operation and further determines that gas turbine engine **202**A should undergo an emergency in-flight restart operation, compressed air may be delivered to ATS **222**A by pneumatic distribution **205** as described above to cause rotation of gas turbine engine **202**A during the emergency in-flight restart. In some examples, as described and illustrated with respect to FIG. **2**, the compressed air may be sourced from second gas turbine engine **202**B, if gas turbine engine **202**B is in operation. In addition to gas turbine engine **202**B, in some examples compressed air delivered to ATS **222**A may include a second source and/or a third source, which may increase the reliability of the emergency in-flight restart operation by increasing from a first source to two sources, and further from two sources to three sources, and/or reduce the time for restart by increasing the force generated by ATS **222**A by expanding more compressed air across ATS **222**A as the number of compressed air sources increases. Any of bleed air from second gas turbine engine **202**B, bleed air from APU **202**C, or compressed air from compressed air tank **242** may be configured as the first source, second source, and or third source of compressed air for pneumatic distribution **205**.

(46) Also illustrated in FIG. **3** is fuel supply system **240**A, which is configured to store and supply

fuel to combustor **212**A of gas turbine engine **202**A for mixing with compressed air from compressor **210**A and combustion. As such, fuel supply may include one or more tanks, valves, pumps, and associated piping. As mentioned above, EEC **203**A may monitor a rotational speed of HP spool **220**A (e.g., via spool speed sensor **221**A). Once the rotational speed of HP spool **220**A reaches a threshold rotational speed for combustion, EEC **203**A may cause fuel to be introduced into combustor **212**A and cause firing of ignitors within combustor **212**A.
(47) In some examples, the relationship between the speed of HP spool **220**A, the introduction of fuel from fuel supply **240**A, the spark for combustion, and other operations of gas turbine engine **202**A may be scheduled as the ignition sequence and stored within processing circuitry of EEC **203**A. During a normal starting operation in which aircraft (**101**, FIG. **1**) is on the ground, the ignition sequence may include EEC **203**A causing fuel to be introduced when HP spool **220**A reaches a first threshold rotational speed, which may be scheduled as the normal starting procedure. In accordance with one or more aspects of this disclosure, the ignition sequence may be modified during an emergency in-flight restart of gas turbine engine **202**A to improve reliability and/or speed of the starting operation. For example, during an emergency in-flight restart, EEC **203**A may cause fuel from fuel supply **240**A to be introduced when HP spool **220**A reaches a second threshold rotational speed. In some examples, the second threshold rotational speed may be lower than the first threshold rotational speed. Put differently, fuel may be introduced earlier in the spool-up process of gas turbine engine **202**A during an emergency in-flight restart procedure than during a normal ground start, which may increase reliability of the start relative to a normal ground start and/or may, in some examples, bring gas turbine engine **202**A up to an operational rotational speed faster than a normal ground start.
(48) However, components of gas turbine engine **202**A may be susceptible to overheating failures if they are allowed to get too hot. Example deleterious effects of overheating gas turbine engine **202**A include cracking or failure of high-temperature coatings on hot section components. Gas turbine engine **202**A may overheat if fuel is injected and combusted in combustor **212**A with insufficient compressed air from compressor **210**A. A critical turbine temperature may be a turbine temperature above which components of gas turbine engine **202**A are prone to overheating and the associated problems. In some examples, the critical turbine temperature may be from about 1400 degrees Celsius to about 2,000 degrees Celsius, although other critical temperatures are considered. If the turbine approaches or reaches the critical temperature, EEC **203**A may flag and register the temperature excursion. For instance, EEC **203**A may generate a data record indicating the occurrence of the temperature excursion. In some examples, approaching or exceeding the critical temperature may mean that aircraft **101** must be inspected for overheat damage upon landing. For instance, based on the data record, maintenance personal may perform an inspection of turbine **214**A. In some examples, EEC **203**A may additionally create a data record indicating the duration of the temperature excursion. Maintenance, inspection, or the expected lifetime schedule of one or more components may be adjusted based at least partially on the duration of time that the temperature excursion lasted.
(49) The first threshold rotational speed, at which fuel is introduced to combustor **212**A when aircraft **101** is on the ground, may be selected at least in part to protect gas turbine engine **202**A from the deleterious effects of overheating. For example, since a faster rotational speed of HP spool **220**A may result in more compressed air moving through combustor **212**A, a first threshold rotational speed may result in a first turbine temperature measured at turbine **214**A. The first threshold rotational speed may be chosen such that the resulting first turbine temperature is below (e.g., with a safety margin) the critical turbine temperature.
(50) As discussed above, fuel may be introduced during an emergency in-flight restart at a second threshold rotational speed of HP spool **220**A, which may be lower than the first threshold rotational speed. Accordingly, introduction of fuel at the second threshold rotational speed may result in a second turbine temperature, which may be higher than the first turbine temperature. In some

examples, the second turbine temperature may be above the critical turbine temperature. As such, the modification to the ignition sequence during the emergency in-flight restart may reduce the expected lifetime of the engine, by operation at the second turbine temperature. Furthermore, the modification to the ignition sequence may result in increased inspection and/or maintenance upon landing aircraft **101**, because EEC **203**A may generate a data record indicating that the turbine temperature approached or exceeded the critical turbine temperature. In some examples, EEC **203**A may generate the data record based on performance of the restarting operation. However, the increased probability of a successful start of gas turbine engine **202**A and/or a faster emergency in-flight restart procedure may provide benefits that outweigh the costs associated with operation at the second temperature.

(51) In some examples, gas turbine engine **202**A may be configured to idle at an idling rotational speed and cruise at a cruising rotational speed of HP spool **220**A. In some examples, the cruising rotational speed may be in a range of from about 5,000 rpm to about 50,000 rpm. In some examples, the idling rotational speed may be in a range of from about 60% to about 95% of the cruising rotational speed (e.g., about 80% of the cruising rotational speed).

(52) In some examples, the first threshold rotational speed may be set relative to the cruising rotational speed or the idling rotational speed. For example, the first threshold rotational speed may be from about 10% to about 40% of the cruising rotational speed. In some examples, the first threshold rotational speed may be from about 10% to about 40% of the idling rotational speed, such as from about 11% to about 30% of the idling rotational speed. In some examples, the second threshold rotational speed may be at least about 10% lower, such as at least about 15% lower, or at least about 20% lower, than the first threshold rotational speed. As an illustrative example, gas turbine engine **202**A may have a design speed (idling, cruising, or the like) of 10,000 rpm. The first threshold rotational speed, at which fuel is introduced during a normal start, may be 3,000 rpm (30% of the design speed). The second threshold rotational speed, at which fuel is introduced during an emergency in-air restart, may be 1,500 rpm (50% lower than the first threshold rotational speed (i.e. 15% of the design speed)).

(53) In some examples, the fuel flow rate may also be modified in accordance with one or more aspects of the current disclosure. For example, EEC **203** may cause fuel to be introduced to combustor **212**A from fuel supply **240**A at a first initial fuel flow rate during a normal starting operation when the vehicle is on the ground. In some examples, in addition to modifying the timing of the fuel introduction such that fuel is introduced at the second threshold rotational speed, EEC **203**A may modify the fuel flow rate, such that fuel is introduced at a second initial fuel flow rate during an emergency in-flight restart. The second initial fuel flow rate may be different than the first initial fuel flow rate. For examples, the second fuel flow rate may be lower than the first flow rate. In this way, the increase in the turbine temperature may be at least partially offset by introducing less fuel. In some examples, modification of the fuel flow rate may cause the air:fuel ratio to remain substantially similar to the air:fuel ratio during a normal, ground start. In some examples, substantially similar or substantially equal, as used herein, comprise values within 10%, or within 20%, of the stated value.

(54) FIG. **4** is a flow diagram illustrating an example technique for restarting a gas turbine engine in-flight, in accordance with one or more aspects of this disclosure. The technique of FIG. **4** may be performed by one or more components of a system, such as system **100** of FIG. **1** or system **200** of FIGS. **2** and **3**. For purposes of this disclosure, the techniques of FIG. **4** will be discussed with reference to aircraft **101** of FIG. **1** and system **200** of FIGS. **2** and **3**.

(55) The technique of FIG. **4** includes receiving, by controller **206**, an emergency restart command for first gas turbine engine **202**A of plurality of gas turbine engines **202** while aircraft **101** is in-flight (**402**). The technique of FIG. **4** also includes determining, via controller **206**, whether first gas turbine engine **202**A is in operation (**404**). In some examples, the determination may be made at least partially based on a speed sensed at speed sensor **221**A, a temperature sensed at the inlet of

turbine **214**A, a power level generated by power turbine **216**A, or a combination of these.
(56) The technique of FIG. **4** may include executing the emergency in-flight restart responsive to determining that first gas turbine engine **202**A is not in operation. The technique includes determining, via controller **206**, whether at least second gas turbine **202**B or APU **202**C in operation (**406**).
(57) The technique of FIG. **4** further includes, responsive to receiving the emergency restart command and determining that second gas turbine engine **202**B, APU **202**C, or both is in operation and that first gas turbine engine **202**A is not in operation, performing an emergency restart of first gas turbine **202**A by at least simultaneously causing air turbine starter **222**A and electric starter **224** to cause rotation of HP spool **220**A of first gas turbine engine **202**A (**408**). In this way, ATS **222**A may use compressed air sourced external to gas turbine engine **202**A, such as bleed air from second gas turbine engine **202**B, bleed air from gas turbine engine **202**C, compressed air tank **242**, or combinations thereof. In some examples, bleed air from second gas turbine engine **202**B may be a first air source supplied to ATS **222**A, and the technique of FIG. **4** may further include causing delivery of compressed air from a second air source to ATS **222**A. In some examples, the second air source include one or more of compressed air tank **242** or APU **202**C.
(58) In some examples, the technique of FIG. **4** may further include driving propulsor **204**A a spool of gas turbine engine **202**A. In some examples, the technique of FIG. **4** may include introducing fuel to combustor **212**A with fuel supply system **240**A of gas turbine engine **202**A at a first threshold rotational speed during a starting operation in which aircraft **101** is on the ground. The technique may further include introducing fuel to combustor **212**A at a second threshold rotational speed during a starting operation in which aircraft **101** is in-flight. The second threshold rotational speed may be different (e.g., lower) than the first threshold rotational speed. Thus, introducing fuel to combustor **212**A at the first threshold rotational speed may result in a first turbine temperature, and introducing fuel to combustor **212**A at the second threshold rotational speed may result in a second turbine temperature. In some examples, the first turbine temperature may be cooler than the second turbine temperature. In some examples, the second threshold rotational speed is at least 10% lower than the first threshold rotational speed.
(59) In some examples, electric starter **224**A is configured to receive electrical energy from a generator of second gas turbine engine **202**B, a battery, an on-board electrical system, a generator of first gas turbine **202**A (generated via windmilling propulsor **204**A) or combinations thereof. In some examples, electrical distribution **207** may include a combination of these sources. In any event, in some examples, electric starter **224**A may cause rotation of HP spool **220**A throughout the emergency in-flight restart to increase the probability of a successful start (e.g., avoid a hung start).
(60) In some examples, gas turbine engine **202**A may include HP spool **220**A, which both ATS **222**A and electric machine **224**A may cause to rotate during a starting operation. Gas turbine engine **202**A may also include LP spool **218**, which may be connected to propulsor **204**A. The technique of FIG. **4** may include generating electrical power from a windmilling operation of propulsor **204**A. The generated electrical power may be supplied by electrical distribution **207** to electrical machine **224**A and used for performing at least part of the emergency in-flight restart of gas turbine engine **202**A. In some examples, EEC **203**A may command a feathering and/or unfeathering of propulsor **204**A. In some examples, responsive to determining that first gas turbine engine **202**A is not in operation and aircraft **101** is in-flight, controller **206** may command a feathering of propulsor **204**A.
(61) In some examples, the technique of FIG. **4** may include mechanically coupling ATS **222**A and electric starter **224**A to a start gear of starter connection components **230**A. For example, both ATS **222**A and electric starter **224** may be mechanically coupled to a first end of a dual-ended start gear.
(62) FIG. **5** is a flow diagram illustrating an example technique for restarting a gas turbine engine in-flight, in accordance with one or more aspects of this disclosure. The technique of FIG. **5** may be performed by one or more components of a system, such as system **100** of FIG. **1** or system **200** of

FIG. **2**. For purposes of this disclosure, the techniques of FIG. **5** will be discussed with reference to system **200** of FIG. **2**.
(63) The technique of FIG. **5** includes causing, with controller **206**, fuel to be introduced to combustor **212**A with fuel supply system **240**A of gas turbine engine **202**A at a first threshold rotational speed during a starting operation in which aircraft **101** is on the ground (**502**). The technique of FIG. **5** further includes causing, via controller **206**, fuel to be introduced to combustor **212**A at a second threshold rotational speed during a starting operation in which aircraft **101** is in-flight (**504**). The second threshold rotational speed may be different (e.g., lower) than the first threshold rotational speed. Thus, introducing fuel to combustor **212**A at the first threshold rotational speed may result in a first turbine temperature, and introducing fuel to combustor **212**A at the second threshold rotational speed may result in a second turbine temperature. In some examples, the second turbine temperature may be higher than the first turbine temperature. In some examples, the first threshold rotational speed is from about 10% to about 40% of a cruising rotational speed. In some examples, the second threshold rotational speed may be at least about 10% lower than the first threshold rotational speed.
(64) In some examples, the second temperature may be above a critical temperature of the gas turbine engine. Thus, operation of the turbine at the second temperature may reduce an expected lifetime of the gas turbine engine. In some examples, the technique of FIG. **5** may include EEC **203**A generating, based on performance of the emergency in-flight restarting operation, a data record indicating the turbine temperature approached or exceeded the critical turbine temperature. In this way, the execution of the emergency in-flight restart technique may be marked for follow-up inspection and/or maintenance upon landing.
(65) The technique of FIG. **5** may include introducing, by fuel supply system **240**A under the control of EEC **203**A, fuel to combustor **212**A at a first initial fuel flow rate during a normal starting operation when the vehicle is on the ground. During an emergency in-flight restart, fuel from fuel supply system **240**A may be introduced at a second initial fuel flow rate, which may be different than the first initial fuel flow rate. For example, the second fuel flow rate may be lower than the first fuel flow rate.
(66) FIG. **6** is a schematic diagram illustrating further details of example system **600**. System **600** includes a motor controller and electric machines in accordance with one or more aspects of the present disclosure. Controller **606** of FIG. **6** may be an example of controller **206** of FIG. **2** and FIG. **3**. Similarly, electric starters **624**A and **624**B (collectively, “electric starters **624**”) may be examples of electric starters **224** of FIG. **2** and FIG. **3**.
(67) As shown in FIG. **6**, HP driver **636** may include power switches **637** that are configured to generate power signals for operating electric starters **624**. Power switches **637** may receive DC electrical energy **607**, which may represent an output of one or both of an AC/DC converter or a DC/DC converter. Control logic **648** may output signals to gates of switches **637** that cause switches **637** to generate the power signals for operating electric starters **524**. As one example, control logic **648** may output signals to gates of power switches **637** that cause power switches **637** to generate a three-phase high power signal for operating electric starters **624**.
(68) The power signals generated by power switches **637** may be routed to electric starters **624** via multiplexor **644**. In some examples, multiplexor **644** may patch the output of switches **637** to electric starter **624**A, electric starter **624**B, or be open. As such, in some examples, multiplexor **644** may function as a three-way switch.
(69) As shown in FIG. **6**, each of electric starters **624** may include a respective set of windings of windings **625**A and **625**B (collectively, “windings **625**”). In some examples, each of electric starters **624** may each further include one or more power switches **650**A and **650**B, respectively (collectively “starter power switches **650**”). The flow of the power signals generated by power switches **637** and starter power switches **650** through windings **625** may generate magnetic fields, which in turn may induce rotation of rotors of electric starters **624**. As noted above, rotation of the

rotors may be transferred into a starting shaft of gas turbine engines **202** (e.g., HP spools **220**). In this way, switches power switches **637** may generate power signals to cause rotation of HP spools **220** of gas turbine engines **202**, which may eventually result in HP spools **220**A reaching the threshold rotational speed for introduction of fuel and combustion.

(70) As mentioned above, during an emergency in-flight restart operation, electric starters **624** may be configured to cause rotation of HP spool **220**A (FIGS. **2** and **3**). In some examples, to increase the probability of a successful start, electric starters **624** may be configured to cause rotation of HP spool **220**A (FIGS. **2** and **3**) throughout a time period during which the emergency in-flight restart is conducted. The time period may be a duration of time that is from about 1 minute to about 15 minutes long, or from about 2 minutes to about 10 minutes long, or from about 3 minutes to about 7 minutes long, in various examples. During an emergency in-flight restart, power switches **637** may generate the power signal to control electric starters **724** continuously or intermittently throughout the duration of the emergency in-flight restart. Since power switches **637**, in most cases, generate thermal energy along with the electric signal due to imperfect efficiency, power switches **637** may increase in temperature while generating the power signal. In some examples, absent cooling mechanisms, power switches **637** may overheat and fail during the emergency in-flight restart.

(71) In accordance with one or more aspects of this disclosure, controller **606** may be provided a thermal energy management system configured to cool controller **606** (e.g., power switches **637**) during an emergency in-flight restart operation. In some examples, the thermal energy management system may include a closed-loop cooling system. The closed-loop cooling system may include a cooling fluid reservoir configured to contain cooling fluid. The cooling fluid may be configured to receive thermal energy from controller **606** during the in-flight restart operation. As used herein, the term about N minutes may be interpreted to be N minutes+/−10%.

(72) FIG. **7** is a conceptual diagram illustrating example system **700** according to the present disclosure. System **700** of FIG. **7** may be generally described similarly to system **200** of FIGS. **2** and **3**, where similar reference numerals indicate like elements. System **700** differs from system **200** as described below. Furthermore, controller **706** may be an example of controller **206** of FIGS. **2** and **3** and/or controller **606** of FIG. **6**. Similarly, EECs **703** may be examples of EECs **203** of FIGS. **2** and **3**.

(73) System **700**, which may also be referred to as the starter system, includes closed-loop cooling system **770** configured to cool electronic components (e.g., power switches) during in emergency in-flight restart of gas turbine engine **702**A, gas turbine engine **702**B, or both. As illustrated in FIG. **7**, some components of gas turbine engines **702** are omitted for clarity.

(74) Closed-loop cooling system **770** includes cooling reservoir **772**, pump **778**, and jackets **776** at least partially surrounding at least one controller of system **700**. Although each of controller **706**, electric starters **724**, and EECs **703** are illustrated as completely surrounded by jackets **776**A, **776**B, **776**C, **776**D, and **776**E respectively, it is understood that in some examples one or more of jackets **776** may be omitted, or may only partially surround a component (e.g., a motor controller housing). For example, cooling system **770** may, in some examples, only include jacket **776**A surrounding controller **706** and not be configured to circulate cooling fluid to EECs **703** or electric starters **724**. In some examples, electric starts **724** may include other cooling features, as will be further described below. It should be noted that although cooling reservoir **772** is illustrated as separate from jacket **776**A, in some examples cooling reservoir **772** and cooling jacket **776**A may be co-located.

(75) Cooling fluid reservoir **772** is configured to contain cooling fluid, which may be pumped about system **770** by pump **778** and returned to cooling fluid reservoir **772**. In some examples, cooling fluid may traverse closed-loop cooling system as a liquid (only illustrated as broken lines flowing through jackets **776**, which indicate the flow of cooling fluid), but it is understood that a phase change may be possible. In examples where the cooling fluid is a liquid, the cooling fluid

may include a refrigerant (e.g., glycol), or may include engine oil, combinations thereof, or another liquid. In examples where the cooling fluid is an engine or turbine oil, there may be benefits using a liquid used in other systems with the cooling fluid used in closed-loop cooling system **770**. Cooling fluid that includes a liquid (e.g., engine oil) may advantageously eliminate the need for an additional fluid within the engine. Furthermore, in examples where the cooling fluid is engine oil, cooling reservoir **772** may be an engine oil reservoir already present on the aircraft, limiting the additional weight of system **770**. In some examples, the cooling fluid may be fuel taken from a fuel supply system (**240**A, FIG. **2**) and circulated through jackets **776** and returned to the fuel supply system (**240**, FIG. **2**). In some examples, since close-loop cooling system **770** is configured to operate rarely and for a relatively short duration, using fuel or engine oil as the cooling fluid may be acceptable (i.e., not evaporate the cooling fluid) and may allow for reduced weight of closed-loop cooling system **770**.

(76) In some examples, cooling fluid that includes a refrigerant may advantageously remain a liquid throughout a flight (e.g., not freeze or boil). Since closed loop cooling system **770** may only be used during an emergency in-flight restart, and since such situations are rare, it is expected that close-loop cooling system **770** will not be activated for large portions of a flight or even at all during most flights. For this reason, a refrigerant cooling fluid that is resistant to both hot and cold temperature extremes may be desirable.

(77) Closed-loop cooling system **770** may be sized to maintain the temperature of controller **706** and/or other electronic components of system **700** below a critical electronics temperature. The critical electronics temperature, in this case, may be the temperature above which power switches (**637**, FIG. **6**) of controller **706** fail. In some examples, the critical electronics temperatures may be about 150 degrees Celsius, or about 175 degrees Celsius, or about 200 degrees Celsius. Accordingly, cooling fluid system **770** may be sized to include enough cooling fluid to keep power switches (**637**, FIG. **6**) below the critical electronics temperature for the duration of the longest expected emergency in-flight restart operation. In some examples, the emergency in-flight restart may define a maximum amount of thermal energy generated. The maximum amount of thermal energy generated may be based on the output of electrical energy from controller **706** to electric starters **724** and the longest duration of the time period over which the emergency in-flight restart may be performed. In some examples, closed-loop cooling system **770** may contain a volume of cooling fluid that absorbs a maximum amount of thermal energy (e.g., without changing phase, or without over-pressuring closed-loop cooling system **770**). In some examples, the maximum amount of thermal energy generated during the emergency in-flight restart may be substantially equal to the maximum amount of thermal energy that can be absorbed by closed-loop cooling system **770**. In this way, power switches (**637**, FIG. **6**) may be protected while closed-loop cooling system **770** adds minimal weight to the aircraft (**101**, FIG. **1**)

(78) In some examples, closed-loop cooling system **770** may be configured to cool components of first gas-turbine **102**A and second gas turbine engine **102**B. As such, cooling fluid reservoir **772** and/or pump **778** may be located within a tail section of a fuselage of aircraft **101** (FIG. **1**) between gas turbine engines **102** disposed on opposing wings, or on opposing sides of the fuselage (**105**, FIG. **1**). In this way, closed-loop cooling system **770** may be configured to serve each gas turbine engine (**102**A, **102**B, FIG. **1**) of a plurality of gas turbine engine **102**.

(79) Since closed-loop cooling system **770** may only need to receive thermal energy generated during a relatively short time period (e.g., an emergency in-flight restart procedure, closed-loop cooling system **770** may be closed-loop. In some examples, closed-loop cooling system **770** may include a substantially fixed amount of cooling fluid, and not include inlets or outlets for new cooling fluid to be brought in or released from closed-loop cooling system **770** during operation. In this way, the additional weight added by closed-loop cooling system **770** may be minimized while ensuring that the temperature sensitive electronics of the motor controllers (e.g., power switches **637**, FIG. **6**) are maintained below the temperature at which they may fail. In some examples,

closed-loop cooling system **770** may not include a heat exchanger, which may save additional weight.

(80) In some examples, cooling fluid reservoir **772** may be a cooling fluid tank defined by wall **774**. In some examples, wall **774** may be flexible and/or stretchable, such that the volume of cooling fluid reservoir **772** may change in response to thermal expansion of cooling fluid contained by cooling fluid reservoir **772**. In some examples, as thermal energy generated by controller **706** is received by closed-loop cooling system **770**, the cooling fluid may expand. In some examples, at least a portion of wall **774** defining cooling fluid reservoir **772** may include a stretchable polymeric material. In this way, the footprint of closed-loop cooling system **770** may be minimized while the internal cooling fluid is protected from contamination.

(81) In some examples, pump **778** may be controlled by controller **706**, as illustrated by the broken line connecting controller **706** and pump **778** in FIG. **7**. Controller **706** may cause pump **778** to activate from a non-operating condition to an operating condition during the emergency in-flight restart, and thus cycle cooling fluid about closed-loop cooling system **770** during the emergency in-flight restart. For example, controller **706** may activate pump **778** responsive to determining that gas turbine engine **102**A is not in operation. Pump **778** may be any suitable pump, such as a centrifugal pump, gear pump, diaphragm pump, rotary pump, axial flow pump, or the like configured to circulate cooling fluid throughout closed-loop cooling system **770**. In some examples, pump **778** may be driven by a 28-volt power supply, which may draw relatively low power and require only a battery small enough to be added to aircraft **101** (FIG. **1**). In other examples, power for pump **778** may be supplied by aircraft batteries through electrical distribution (**207**, FIG. **2**).

(82) In some examples, controller **706**, electric starters **724**, and/or EECs **703** may include a temperature sensor **779** (e.g., temperature sensor **779**A housed within controller **706**, temperature sensor **779**B housed within electric starter **724**A, temperature sensor **779**C housed within electric starter **724**B, temperature sensor **779**D housed within EEC **703**A, and temperature sensor **779**E housed within EEC **703**B)). In some examples, controller **706** may be configured to activate pump **778** responsive to any one of temperature sensors **779** approaching or exceeding the critical electronics temperature. For example, pump **778** may be activated by controller **706** responsive to any one of temperature sensors **779** reaching a temperature within about 10%, or about 20%, or about 30% of the critical electronics temperature, when measured on an absolute temperature scale such as Kelvin.

(83) FIG. **8** is a conceptual diagram illustrating example system **800** according to the present disclosure. System **800** of FIG. **8** may be generally described similarly to system **700** of FIG. **7**, where similar reference numerals indicate like elements. System **800** differs from system **700** as described below. System **800** includes closed-loop cooling system **870** configured to cool electronic components (e.g., power switches) during in emergency in-flight restart of gas turbine engine **702**A, gas turbine engine **702**B, or both.

(84) Closed-loop cooling system **870** includes cooling reservoir **872**, pump **878**, and jacket **876**A surrounding controller **806**, such that cooling fluid may flow from cooling reservoir **872** defined by wall **874** to jacket **876**A to liquid-cool controller **806**. As such, controller **806**, which may include power switches and be subject to a high thermal load during an emergency restart operation, may be cooled by closed-loop cooling system **870** during the emergency in-air restart to maintain controller **806** below a critical electronics temperature. In some examples, as illustrated, closed-loop cooling system **870** may liquid cool critical electronics components while other components, such as EEC **803**A, EEC **803**B, electric machine **824**A, and/or electric machine **824**B, which may not be subject to such a high thermal load during the emergency in-air restart and, may be air-cooled or cooled by another system apart from closed-loop cooling system **872**.

(85) FIG. **9** is a conceptual cross-section diagram of an example electronic machine **924** of system **900**, according to one or more aspects of the present disclosure. Electric machine **924** may be an

example of electric starters **224** of FIGS. **2** and **3**, EECs **203** of FIGS. **2** and **3**, or controller **206** of FIGS. **2** and **3**. Electric machine **924** is contained within housing **980**, with internal components of electric machine **924** omitted for clarity. As mentioned above, in some examples housing **980** may be partially or totally surrounded by jacket **876**, such that electric machine **924** may be liquid-cooled by cooling fluid flowing through jacket **976**. However, jacket **976** is optional, and housing **980** does not necessarily include a portion of a closed-loop cooling system (**770**, FIG. **7**). In some examples, to remove thermal energy from electric machine **924**, housing **980** or a jacket **976** connected thereto may include one or more cooling features configured remove thermal energy from the starter system through convective heat transfer during the emergency in-flight restart. For example, jacket **976** and/or controller housing **980** may include a plurality of fins **982**A, **982**B, **982**C (collectively "fins **982**"). Fins **982** may transfer thermal energy generated by electric machine **924** more efficiently and/or quickly to the surroundings of electrical machine **924** than a housing **980** or jacket **976** that does not include fins **982** by increasing the surface area available for heat transfer to the environment.

(86) Since electric machine **924** may be located in a cramped spaced in the tail section of a fuselage (**105**, FIG. **1**), fins **982** may be small relative to a maximum cross-sectional distance of housing **980**. For example, fins **982** may protrude a distance P from housing **980**. Housing **980** may define a maximum cross-sectional distance D. In some examples, P/D may be less than or equal to about 0.5, or less than or equal to about 0.25, or less than or equal to about 0.1 Fins **982** may be relatively small when compared to conventional heat sink fins, as described, by taking advantage of the natural flow of air in the tail-section of the fuselage, which aids in removing thermal energy. In this way, the footprint of electric machine **924** may be minimized without overheating. In some examples, to further bolster the heat transfer accomplished by fins **982**, a portion of one or more surfaces of fins **882** may include a phase-change material (not illustrated). An example phase-change material includes paraffin wax. Phase-change materials such as paraffin wax may improve cooling by continuing to absorb heat from housing **980** throughout the phase change process.

(87) FIG. **10** is a conceptual cross-section diagram of example electronic machine **1024** of system **1000**, according to one or more aspects of the present disclosure. Electric machine **1024** may be an example of electric starters **224** of FIGS. **2** and **3**, EECs **203** of FIGS. **2** and **3**, or controller **206** of FIGS. **2** and **3**. Electric machine **1024** is contained within housing **1080**, with internal components of electric machine **1024** omitted for clarity. As mentioned above, in some examples electric machine **1024** may be air-cooled. In such examples, a closed-loop cooling system may not be configured to partially surround housing **1080**. In such examples, housing **1080** may include one or more cooling features configured remove thermal energy from the starter system during the emergency in-flight restart. For example, housing **1080** may include a plurality of fins **1082**A, **1082**B, **1082**C (collectively "fins **1082**"). Fins **1082** may transfer thermal energy generated by electric machine **924** more efficiently and/or quickly to the surroundings of electrical machine **924** than a housing **980** or jacket **976** that does not include fins **982** by increasing the surface area available for heat transfer to the environment.

(88) FIG. **11** is a conceptual cross-section diagram of example electronic machine **1124** of system **1100**, according to one or more aspects of the present disclosure. Electric machine **1124** may be generally similarly to electric machine **1024** of FIG. **10**, differing as described below. In some examples, each of fins **1182**A, **1182**B, and **1182**C may be a hollow fin which includes a respective chamber of chambers **1184**A, **1184**B, and **1184**C. In some examples, chamber **1184**A may be at least partially filled with a phase change material (e.g., paraffin wax). Hollow fins **1182** which include phase change material may help to maintain electric machine **1124** within housing **1180** below a critical electronics temperature. Furthermore, the phase change material may be encapsulated within chambers **1184** and prevented from becoming a debris risk if it were to liquify or gasify.

(89) FIG. **12** is a flow diagram illustrating an example technique for managing thermal energy

during an in-flight restart, in accordance with one or more examples of the present disclosure. The technique of FIG. **12** may be performed by one or more components of a system, such as system **100** of FIG. **1** or system **200** of FIGS. **2** and **3**, system **600** of FIG. **6**, system **700** of FIG. **7**, or system **800** of FIG. **8**. For purposes of this disclosure, the techniques of FIG. **9** will be discussed with reference to system **700** of FIG. **7** and system **800** of FIG. **8**.

(90) The technique of FIG. **12** includes starting first gas turbine engine **102**A, which is configured to provide propulsion to aircraft **101** while aircraft **101** is in-flight, by controlling motor controller **606** (**1202**). Motor controller **606** includes power switches **637**. Power switches **637** are configured to generate a power signal to control an electric motor **724** to cause rotation of HP spool **220**A to start gas turbine engine **102**A. However, power switches **637** may overheat and fail should the temperature of power switches **637** approach or exceed a critical electronics temperature. Since an emergency in-flight restart may take several tries or extend over several minutes, during which time controller **606** and electric motors **724** are generating thermal energy, power switches **637** may be prone to overheat during an emergency in-flight restart if not properly cooled. However, typical cooling systems may undesirably add weight to aircraft **101**.

(91) In accordance with one or more aspects of the disclosure, the technique of FIG. **12** includes cooling controller **706** during the in-flight restart with closed-loop cooling system **770** (**1204**). Closed-loop cooling system **770** includes cooling fluid reservoir **772** containing cooling fluid. The cooling fluid is configured to receive thermal energy from controller **706** during the in-flight restart operation. In some examples, cooling system **770** may be further configured to cool electric starters **724** and/or EECs **703** during the emergency in-flight restart.

(92) In some examples, closed-loop cooling system **770** may be sized to maintain the temperature of controller **706** and/or EECs **703** below a threshold temperature for a time period during which the in-flight restart operation is conducted. The threshold temperature may be a critical electronics temperature, above which power switches **637** within controller **706** and/or EECs **703** may overheat and fail. In some examples, the critical electronics temperature may be about 150 degrees Celsius, or about 175 degrees Celsius, or about 200 degrees Celsius. In some examples, the time period may be from about 1 minute to about 15 minutes in duration, or from about 2 minutes to about 10 minutes in duration, or from about 3 minutes to about 7 minutes in duration. In this way, the volume of cooling fluid (e.g., the volume of cooling fluid reservoir **772**) may be selected based on the amount of cooling required to keep the temperature below the critical electronics temperature for the maximum duration of time that it may take to restart gas turbine engine **202**A.

(93) In some examples, closed-loop cooling system **770** may add as little weight to aircraft **101** as possible while maintaining power switches **637** below the critical electronics temperature. As such, closed-loop cooling system may be located centrally to reduce required piping. For example, aircraft **101** may include gas turbine engine **102** disposed on opposing sides of fuselage **105**. Closed-loop cooling system **770** may be disposed within a tail section of a fuselage disposed between opposing wings of aircraft **101**. In some examples, closed-loop cooling system **770** may not include a heat exchanger configured to transfer the thermal energy generated by controller **706** outside of closed-loop cooling system **770**, because closed-loop cooling system **770** need only be activated during an emergency in-flight restart, which occurs over a relatively short time period. In this way, the weight added to aircraft **101** by closed-loop cooling system **770** may be minimized by omission of unnecessary equipment.

(94) In some examples, the technique of FIG. **12** may include activating closed-loop cooling system **770** from a non-operating condition to an operating condition during an emergency in-flight restart operation of gas turbine engine **702**A. For example, EEC **703**A may sense, (e.g., based on a speed, a temperature, a power level, etc. of gas turbine engine **702**A) that gas turbine **702**A is not in operation. Responsive to determining that gas turbine engine **702**A is not in operation, EEC **703**A may output a signal to controller **706**. Responsive to receiving the signal, controller **706** may activate cooling system **770** (e.g., by causing power to be supplied to pump **778**). In some

examples, cooling system **770** may be activated responsive to controller **706** determining, based at least partially on a temperature sensed at a temperature sensor **779**, that a threshold temperature (e.g., a temperature approaching or exceeding a critical electronics temperature) has been reached in one or more of controller **706**, EECs **703**, or electric starters **724**.

(95) In some examples one or more of controller **706**, EECs **703**, and/or electric starters **724** may include cooling features configured to emit thermal energy from the starter system into the surroundings, further cooling the starter system. For example, electric machine **824** may include controller housing **880**, which may be at least partially surrounded by jacket **876**. Jacket **876** may contain cooling fluid, which may be liquid. In some examples, housing **880**, jacket **876**, or both may define plurality of fins **882**, which may increase the surface area for heat transfer relative to a housing **880** or jacket **876** that does not define cooling features. In some examples, which reference to FIG. **11**, fins **1182** may be hollow. Hollow fin **1182** may define chamber **1184**, which may be at least partially filled with phase change material.

(96) The following numbered examples demonstrate one or more aspects of the disclosure.

(97) Example 1: A system includes a gas turbine engine configured to provide propulsion to an aircraft; a starter system configured to start the gas turbine engine, wherein the starter system comprises a motor controller; and a closed-loop cooling system configured to cool the motor controller during an emergency in-flight restart operation of the gas turbine engine, wherein the closed-loop cooling system includes a cooling fluid reservoir containing cooling fluid, wherein the cooling fluid is configured to receive thermal energy from the motor controller during the emergency in-flight restart operation of the gas turbine engine.

(98) Example 2: The system of example 1, wherein: the motor controller includes power switches, the power switches being configured to generate a power signal to control an electric machine configured to start the gas turbine engine, the power switches generate heat while generating the power signal, and the closed-loop cooling system is configured to cool the power switches during the emergency in-flight restart.

(99) Example 3: The system of any of examples 1 and 2, wherein: the closed-loop cooling system is sized to absorb a maximum amount of thermal energy, and wherein the maximum amount of thermal energy is substantially equal to the maximum amount of thermal energy that can be generated during the emergency in-flight restart.

(100) Example 4: The system of any of examples 2 and 3, wherein: the closed-loop cooling system is configured to keep the power switches at or below a critical electronics temperature during the emergency in-flight restart, and the critical electronics temperature is about 175 degrees Celsius.

(101) Example 5: The system of any of examples 3 and 4, wherein a time period during which the emergency in-flight restart is conducted is from about 1 minute to about 15 minutes in duration.

(102) Example 6: The system of any of examples 1 through 5, wherein the gas turbine engine is a first gas turbine engine attached to a first side of a fuselage of the aircraft, wherein the system further comprises a second gas turbine engine disposed on a second side of the fuselage opposite the first side, and wherein the closed-loop colling system is disposed within a tail section of the fuselage.

(103) Example 7: The system of any of examples 1 through 6, wherein the closed-loop cooling system does not include a heat exchanger configured to transfer the thermal energy generated by the motor controller outside of the closed-loop cooling system.

(104) Example 8: The system of any of examples 1 through 7, wherein the closed-loop cooling system comprises a liquid cooling fluid, and wherein the closed-loop cooling system includes a cooling fluid pump.

(105) Example 9: The system of example 8, wherein the liquid cooling fluid includes at least one of glycol or engine oil.

(106) Example 10: The system of any of examples 8 and 9, wherein the cooling fluid pump is driven by a 28-volt power supply.

(107) Example 11: The system of any of examples 1 through 10, wherein at least a portion of a wall of the cooling fluid reservoir is configured to flex to change shape.
(108) Example 12: The system of any of examples 8 through 11, further comprising a controller configured to activate the closed-loop cooling system from a non-operating condition to an operating condition during the restart operation of the gas turbine engine.
(109) Example 13: The system of example 12, further comprising a temperature sensor, wherein the controller is configured to activate the closed-loop cooling system based at least partially on the temperature sensor sensing a critical electronics temperature.
(110) Example 14: The system of any of examples 12 and 13, wherein the controller is configured to activate the closed-loop cooling system responsive to determining that the gas turbine engine is not in operation.
(111) Example 15: The system of any of examples 12 through 14, wherein the motor comprises one or more cooling fins to remove thermal energy from the starter system through convective heat transfer.
(112) Example 16: The system of example 15, wherein at least one cooling fin of the one or more cooling fins comprises a phase change material on at least one surface of the fin, the phase change material configured to absorb thermal energy by changing phases from a solid to a liquid during the emergency in-flight restart.
(113) Example 17: The system of any of examples 2 through 16, wherein the motor is at least partially surrounded by a jacket configured to contain the cooling fluid.
(114) Example 18: A method includes starting a gas turbine engine configured to provide propulsion to an aircraft while the aircraft is in-flight by controlling a motor controller; and cooling the motor controller with a closed-loop cooling system during an emergency in-flight restart operation of the gas turbine engine, wherein the closed-loop cooling system includes a cooling fluid reservoir containing cooling fluid, wherein the cooling fluid is configured to receive thermal energy from the motor controller during the emergency in-flight restart operation of the gas turbine engine.
(115) Example 19: The method of example 18, further comprising; generating, by power switches of the motor controller, a power signal to control an electric machine to start the gas turbine engine, wherein the power switches increase in temperature while generating the power signal, and wherein the closed-loop cooling system cools the power switches during the emergency in-flight restart.
(116) Example 20: The method of any of examples 18 and 19, wherein cooling the motor controller during the emergency in-flight restart operation comprises cooling the motor controller for a time period that is from about 1 minute to about 15 minutes in duration.
(117) Example 21: The method of any of examples 18 through 20, further comprising activating the closed-loop cooling system from a non-operating condition to an operating condition during the restart operation of the gas turbine engine.
(118) Example 22: The method of example 21, wherein activating the closed-loop cooling system is based at least partially on a temperature sensor sensing a critical electronics temperature.
(119) Example 23: The method of any of examples 18 through 22, further comprising activating the closed-loop cooling system responsive to determining that the gas turbine engine is not in operation.
(120) Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. A system comprising: a gas turbine engine configured to provide propulsion to an aircraft; a starter system configured to start the gas turbine engine, wherein the starter system comprises a motor controller; and a closed-loop cooling system configured to cool the motor controller during

an emergency in-flight restart operation of the gas turbine engine, wherein the closed-loop cooling system includes a cooling fluid reservoir containing cooling fluid, wherein the cooling fluid is configured to receive thermal energy from the motor controller during the emergency in-flight restart operation of the gas turbine engine, wherein the closed-loop cooling system does not include a heat exchanger configured to transfer thermal energy generated by the motor controller outside of the closed-loop cooling system.
2. The system of claim 1, wherein: the motor controller includes power switches, the power switches being configured to generate a power signal to control an electric machine configured to start the gas turbine engine, the power switches generate heat while generating the power signal, and the closed-loop cooling system is configured to cool the power switches during the emergency in-flight restart.
3. The system of claim 1, wherein: the closed-loop cooling system is sized to absorb a maximum amount of thermal energy, and wherein the maximum amount of thermal energy is substantially equal to the maximum amount of thermal energy that can be generated during the emergency in-flight restart.
4. The system of claim 2, wherein: the closed-loop cooling system is configured to keep the power switches at or below a critical electronics temperature during the emergency in-flight restart, and the critical electronics temperature is about 175 degrees Celsius.
5. The system of claim 3, wherein a time period during which the emergency in-flight restart is conducted is from about 1 minute to about 15 minutes in duration.
6. The system of claim 1, wherein the closed-loop cooling system comprises a liquid cooling fluid, and wherein the closed-loop cooling system includes a cooling fluid pump.
7. The system of claim 6, wherein the liquid cooling fluid includes at least one of glycol or engine oil.
8. The system of claim 6, wherein the cooling fluid pump is driven by a **28-** volt power supply.
9. The system of claim 1, wherein at least a portion of a wall of the cooling fluid reservoir is configured to flex to change shape.
10. The system of claim 6, further comprising a controller configured to activate the closed-loop cooling system from a non-operating condition to an operating condition during the emergency in-flight restart operation of the gas turbine engine.
11. The system of claim 10, further comprising a temperature sensor, wherein the controller is configured to activate the closed-loop cooling system based at least partially on the temperature sensor sensing a critical electronics temperature.
12. The system of claim 10, wherein the controller is configured to activate the closed-loop cooling system responsive to determining that the gas turbine engine is not in operation.
13. The system of claim 10, wherein the motor controller comprises one or more cooling fins to remove thermal energy from the system through convective heat transfer.
14. The system of claim 13, wherein at least one cooling fin of the one or more cooling fins comprises a phase change material on at least one surface of the fin, the phase change material configured to absorb thermal energy by changing phases from a solid to a liquid during the emergency in-flight restart.
15. The system of claim 2, wherein the motor controller is at least partially surrounded by a jacket configured to contain the cooling fluid.
16. A method comprising: starting a gas turbine engine configured to provide propulsion to an aircraft while the aircraft is in-flight by controlling a motor controller; and cooling the motor controller with a closed-loop cooling system during an emergency in-flight restart operation of the gas turbine engine, wherein the closed-loop cooling system includes a cooling fluid reservoir containing cooling fluid, wherein the cooling fluid is configured to receive thermal energy from the motor controller during the emergency in-flight restart operation of the gas turbine engine, and wherein the closed-loop cooling system does not include a heat exchanger configured to transfer

thermal energy generated by the motor controller outside of the closed-loop cooling system.
17. The method of claim 16, further comprising; generating, by power switches of the motor controller, a power signal to control an electric machine to start the gas turbine engine, wherein the power switches increase in temperature while generating the power signal, and wherein the closed-loop cooling system cools the power switches during the emergency in-flight restart.
18. The method of claim 16, wherein cooling the motor controller during the emergency in-flight restart operation comprises cooling the motor controller for a time period that is from about 1 minute to about 15 minutes in duration.
19. The system of claim 1, wherein the cooling fluid is configured to remain a liquid throughout a flight.